

Punto 1

TOKENS = [

IDENTIFIER: $r'[a-zA-Z][a-zA-Z0-9]^*$

integer literal $r'[0-9]^+$

floating point $r'[0-9]^+([eE][+-]?[0-9]^+)?[.][0-9]^+[eE][+-]?[0-9]^+$

string literal $r'\"([^\"]|\\\"|\\\\)^*\"'$

white space $r'[\t+]^*$

operator $r'[\+\-*/=]$

keyword $r'(\text{if}|\text{else}|\text{while}|\text{return})'$

hexadecimal $r'0[xX][0-9a-fA-F]^+'$

semicolon $r';'$

Punto 2

a. $x = 5 + 3 * 2;$

S
|
Program
|
Statements
|
Assignment

S
|
Program
|
Statement list
|
Statement Statement list
|
Assignment <lambda>
|
Identifier "=" Expression ";"
|
x Term expression

Factor Term "+" Term expression
| |
Number <lambda> Number Term expression
| | |
5 3 "*" Factor Term
| |
Number <lambda>
|
2

```

6.  if (x > 0) {
    y = x - 1;
  } else {
    y = 0;
  }

```

```

S
|
Program
|
Statement list
|
Statement Statement list
|
do statement

```

"if" "(" Expression ")" "{" Statement list "}" Else Part

```

      Term Expression
      |               |
Factor Term  ">" Term expression
|           |               |
Identifier  "<lambda>" Factor Term  "<lambda>"
|           |               |
x           0               0

```

```

Assignment
|
Identifier "=" Expression ";"
|           |               |
y           Term expression
|           |               |
Factor Term
|           |               |
Identifier "<lambda>"
|           |               |
x           "-" Term Expression
|           |               |
Factor Term  "<lambda>"
|           |               |
Number  "<lambda>"
|           |               |
0           1

```

— Else part

"else" "{" Statement list "}"

```

Assignment
|
Identifier "=" Expression ";"
|           |               |
y           Term Expression
|           |               |
Factor Term  "<lambda>"
|           |               |
Number  "<lambda>"
|           |               |
0           0

```


c. while (x < 10) {

x = x + 1;

}

S

|

Program

|

Statement List

|

Statement Statement List - <lambda>

|

while Statement

|

while "(" Expression ")" "{" Statement List "}"

Term Expression

Factor Term

|

Identifier

|

x

<lambda>

"<" Term Expression

|

Factor Term

|

Number

|

10

<lambda>

<lambda>

— Statement List

|

Assignment "=" Expression ";"

|

x

Term Expression

Factor Term

|

Identifier

|

x

<lambda>

"+" Term Expression

|

Factor Term

|

Number

|

1

<lambda>

<lambda>

4. `return (a+b)*c;`

S

Program

Statement List

Statement Statement List $\rightarrow$ $\langle \text{lambda} \rangle$

Return Statement

'return' Expression ';'

Term expression

Factor Term $\langle \text{lambda} \rangle$

"(" Expression ")" "*" Factor Term $\langle \text{lambda} \rangle$

Term expression

Identifier

Factor Term

"+" Term expression "c"

Identifier $\langle \text{lambda} \rangle$

$\langle \text{lambda} \rangle$

a

Factor Term

Identifier $\langle \text{lambda} \rangle$

b